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Real-Time Weapon Detection in Surveillance Video Footages

Universidade de Aveiro - DETI

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Topics

- Context and Motivation
- Related Work
- Overview of Object Detection, Algorithms and Performance Metrics
- CCTV System Setup and Configuration
- Dataset Sources
- High Level Architecture
- Preliminary Results
- Future Work

Context and Motivation

- Rising Crime Rates Despite Strict Weapon Laws
- Increased Reliance on CCTV Systems
- Limitation of Human-Monitored CCTV Systems
- **Goal:** Automated Weapon Detection System

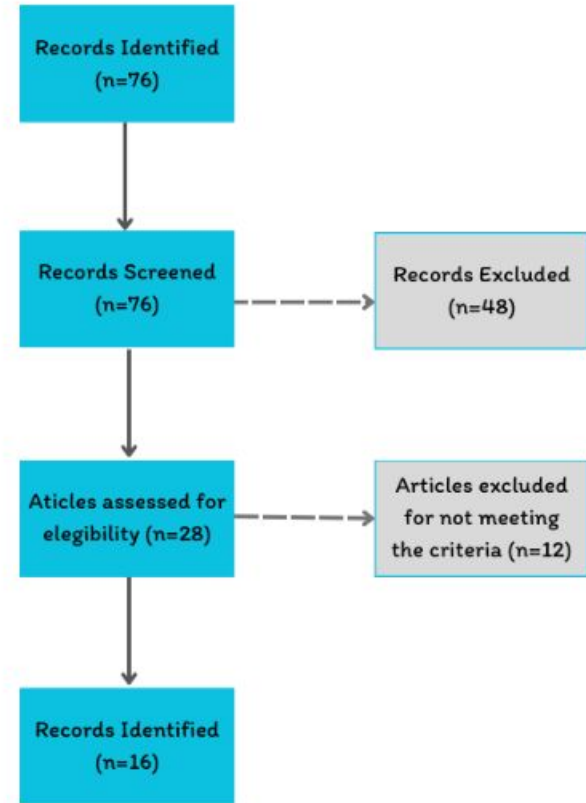
Participações registadas por distrito e região autónoma											
	Distritos	Ano 2021	Ano 2022	Var %	Dif		Distritos	Ano 2021	Ano 2022	Var %	Dif
✖	Aveiro	400	470	17,5 %	70	🟡	Portalegre	93	93	0,0 %	0
✖	Beja	84	120	42,9 %	36	✖	Porto	1 886	2 100	11,3 %	214
✔	Braga	469	445	-5,1 %	-24	✖	Santarém	314	375	19,4 %	61
✔	Bragança	56	46	-17,9 %	-10	✖	Setúbal	1 234	1 330	7,8 %	96
✖	Castelo Branco	106	148	39,6 %	42	✖	Viana do Castelo	102	124	21,6 %	22
✖	Coimbra	252	317	25,8 %	65	🟢	Vila Real	100	90	-10,0 %	-10
✖	Évora	93	136	46,2 %	43	✖	Viseu	153	158	3,3 %	5
✖	Faro	707	876	23,9 %	169	✖	Madeira	177	269	52,0 %	92
✖	Guarda	56	60	7,1 %	4	✖	Açores	220	232	5,5 %	12
✖	Leiria	286	344	20,3 %	58	✖	Não especificado	207	402	94,2 %	195
✖	Lisboa	4 619	5 146	11,4 %	527	✖	Total	11 614	13 281	14,4 %	1 667

Geographical Distribution of Crime Rate Increase
in Portugal by District in 2022, from RASI

Related Work

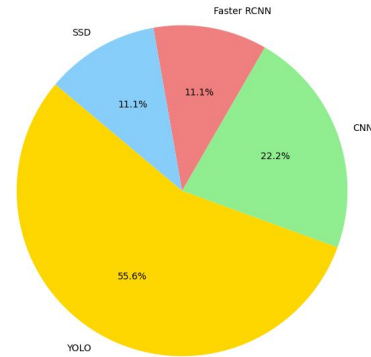
- Search applied over the following research platforms
 - ResearchGate
 - Scopus
 - arXiv
- Strategic Keywords Selection
- Emphasis on Recent Publications
- **Example Query:**

“Deep” AND “Learning” AND “Real” AND “Time”
AND “Weapon” AND “Detection”

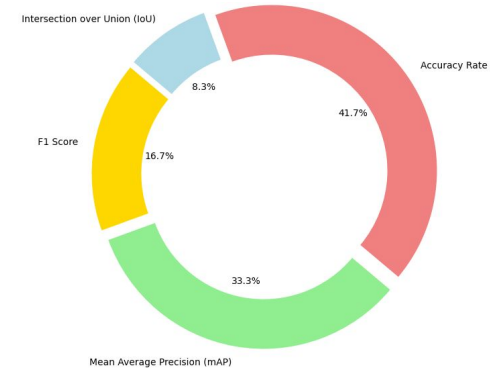


Related Work

- Overview of Pre-processing Techniques
- Detailed Analysis of Weapon Detection Methods
- Results and Performance Analysis
- Limitations in the Methodologies



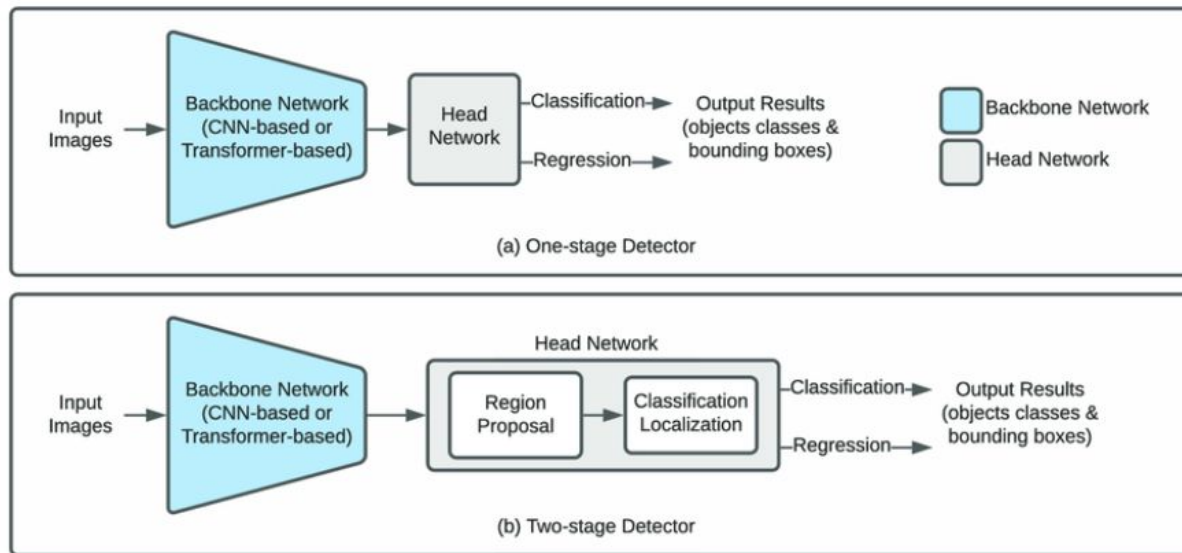
Distribution of algorithms used in studies



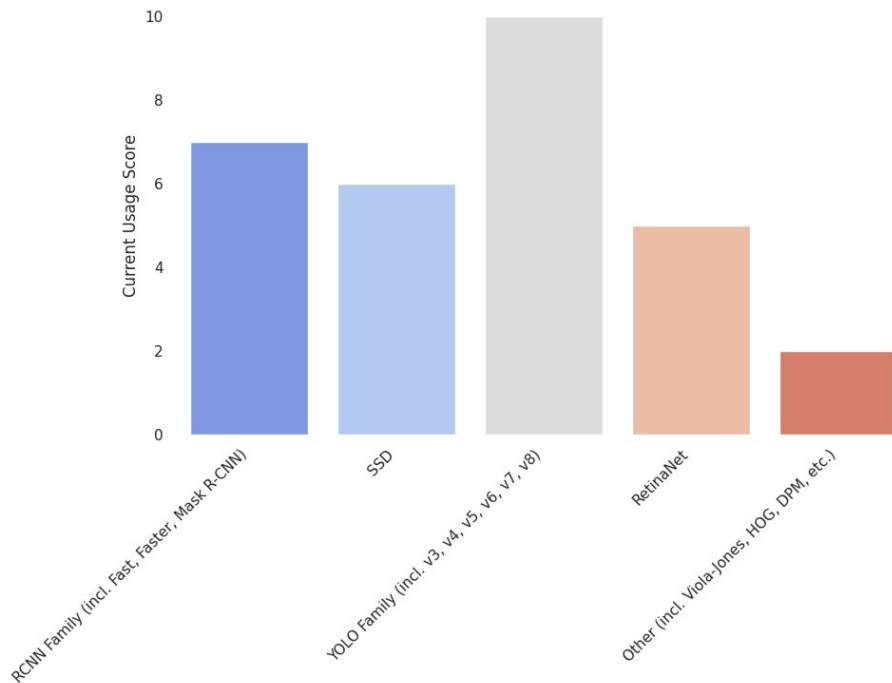
Distribution of metrics used in studies

Overview of Object Detection

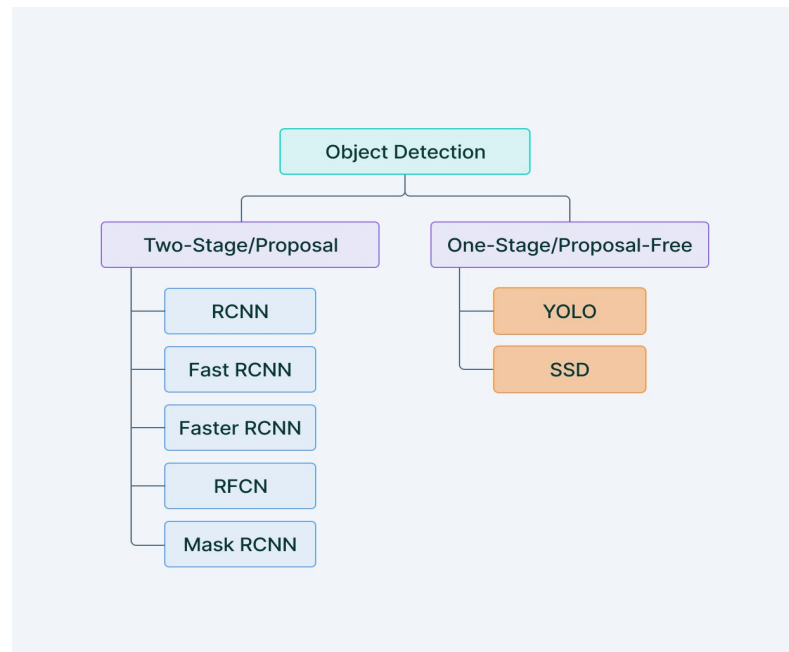
- Aims to Locate and Classify Objects in Images
- Two-stage detectors vs One-stage detectors



Object Detection Algorithms Overview



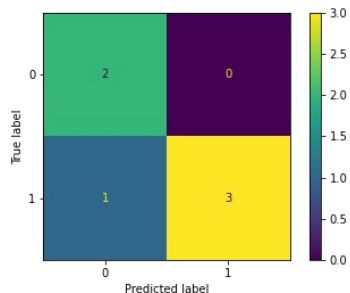
Comparative Usage of Object Detection Algorithms



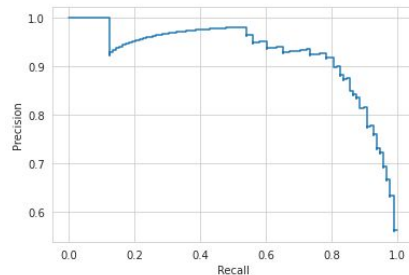
Two Stage vs One Stage Detectors

Object Detection Performance Metrics

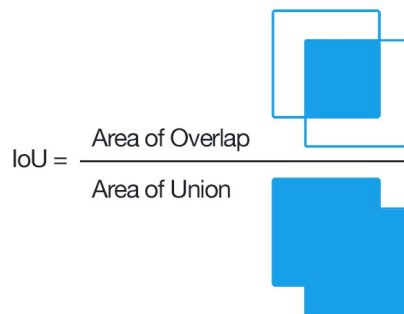
- Accuracy
- Recall
- F1 Score
- Average Precision and Mean Average Precision
- Intersection over Union (IoU)



Confusion Matrix



Precision Recall Curve



CCTV System Setup and Configuration

- Key Factors for Effective CCTV Security System
- Influence of Camera Design on Crime Prevention
 - Semi-covert Dome vs Covert Cameras
- Video Quality Concerns



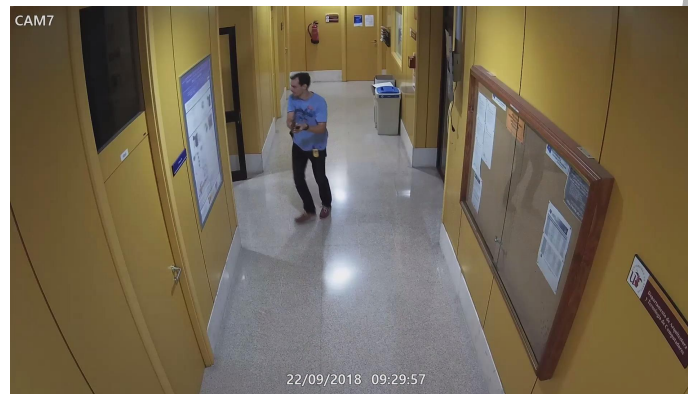
Semi-covert dome camera



Covert camera

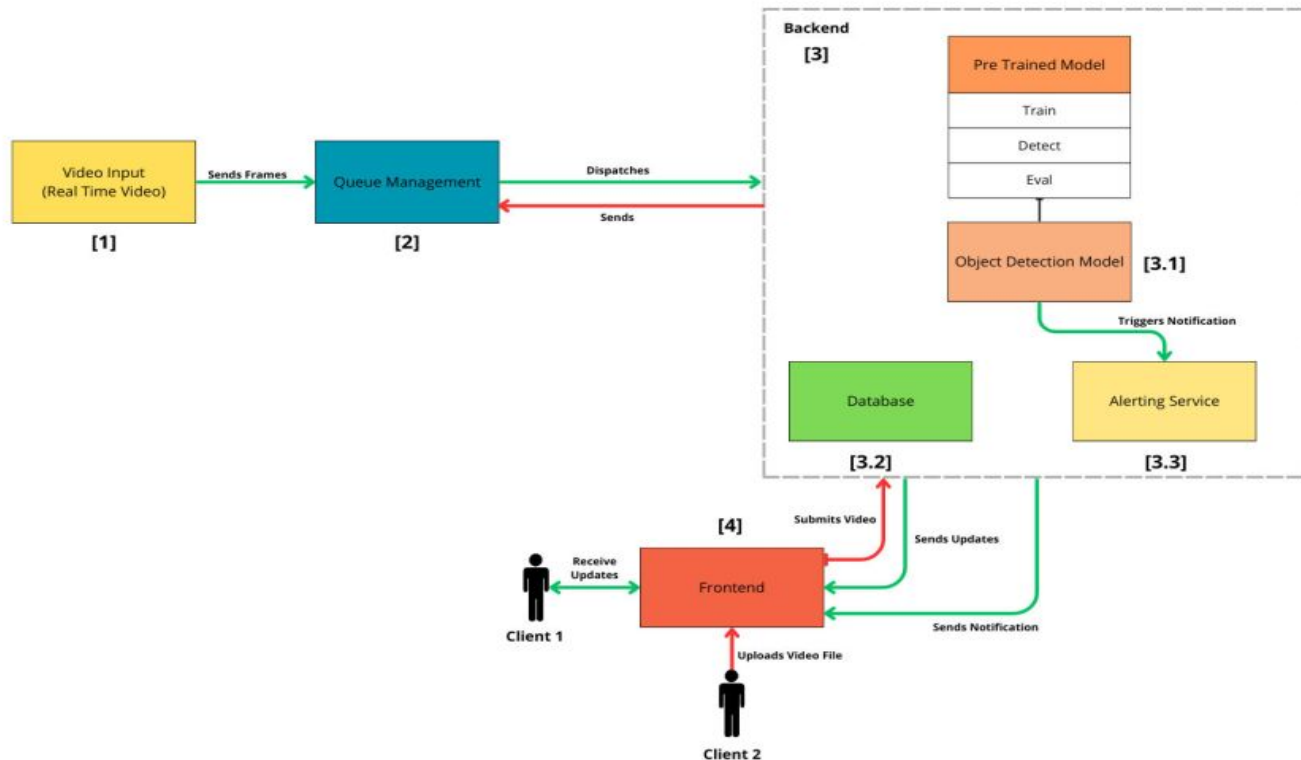
Dataset Sources

- DaSCI
- Mock Attack Dataset
- Total: 11020 Images



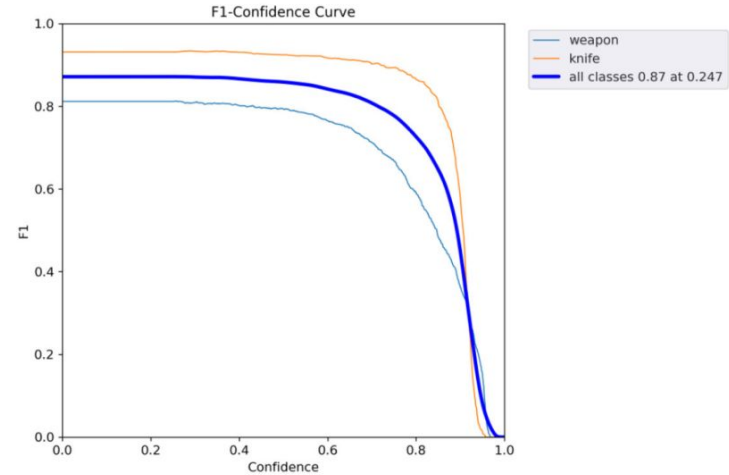
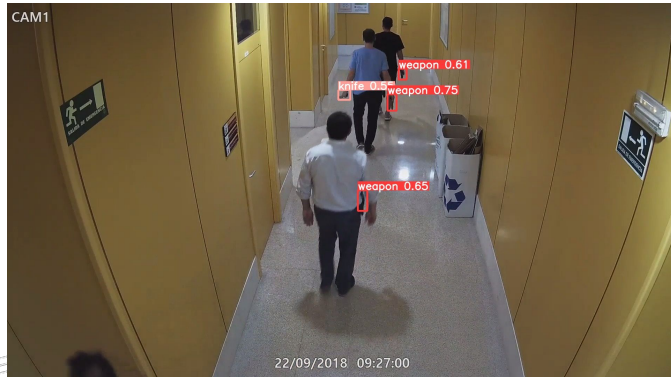
Source	Images	Description
DaSCI	5371	Variety of weapons displayed in different positions
DaSCI	500	Random everyday objects
Mock Attack Dataset	607	Cam1: located in one of two corridors, with uniform lighting and some conflicting objects such as doors or bins
Mock Attack Dataset	3511	Cam5: at the entrance of a university module, features objects like a black carpet on the floor, with irregular lighting.
Mock Attack Dataset	1031	Cam7: in the other corridor, similar to Cam1 but with more conflicting objects

High Level Architecture



Preliminary Results

- Automated Parameter Selection Script for YOLOv5
- Train and Test with the Created Dataset
- 80% for Training, 20% for Testing
- F1 score of 0.87 and Accuracy of 0.9



F1-score results with 100 epochs and using a batch size of 64

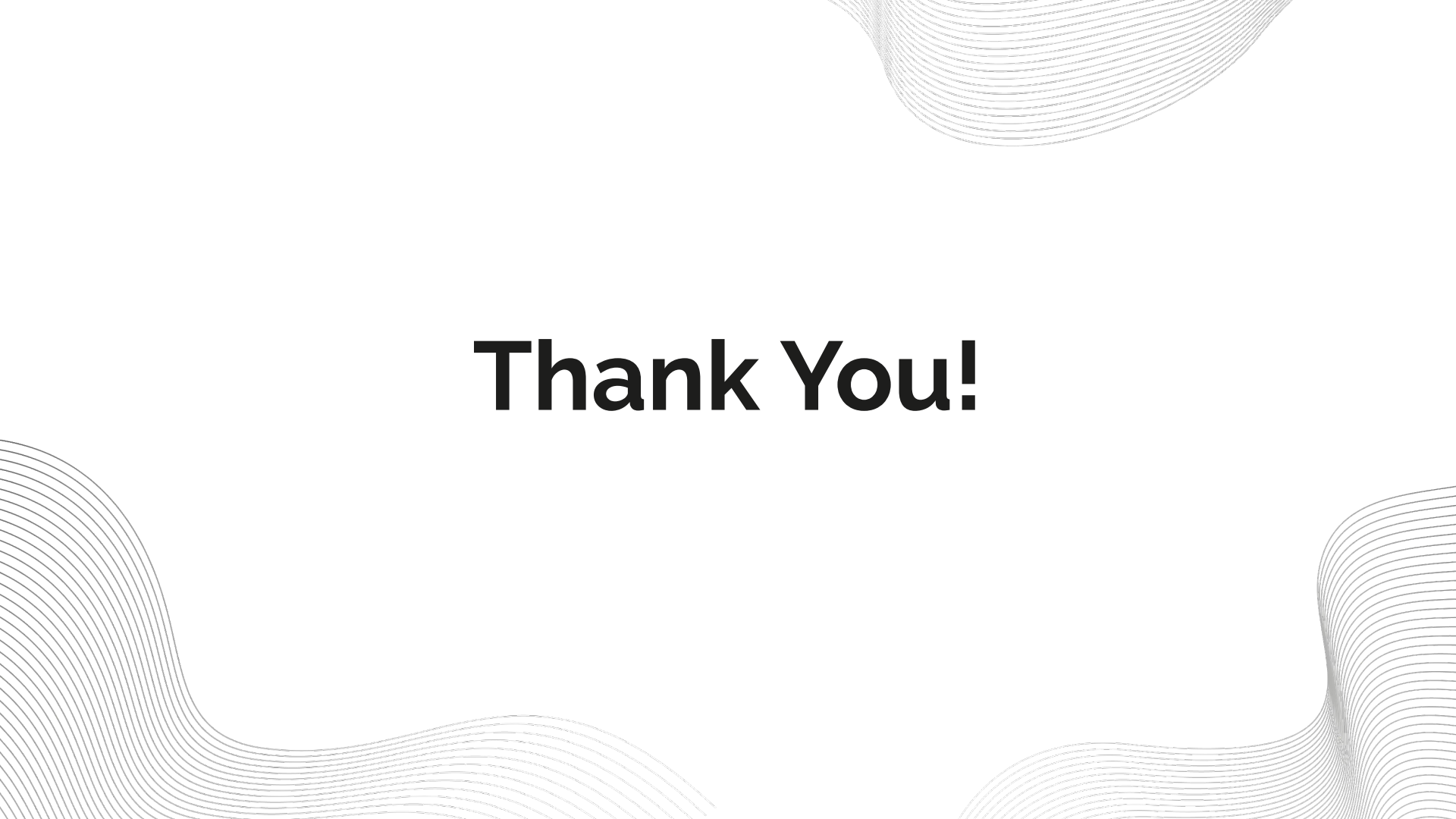
Work Plan

Task Name	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
Thesis Writing - State of Art										
Automated Hyperparameter Tuning Script Development										
Architecture Planning										
Dataset Enrichment										
Thesis Writing - Introduction and Methodology										
Train and Test Model										
Pre Thesis Completion and Review										
Explore and Choose Technologies										
API Endpoints Development										
Frontend Design and Development										
Alerting Service Development										
Application Modules Integration										
Thesis Writing - Proposal Chapter										
Test model with real-time video data										
Identify potential limitations in diverse conditions										
Peer review and supervisor feedback										
Draft conclusions and recommendations for future work										
Deploy the model in a controlled environment										
Scalability Analysis and Optimization										
Thesis Writing - Results, Conclusion and Future Work										

Future Work

- Technology Analysis and Selection
- Development of Architecture Modules
- Enhance Dataset Division
- Test Multiple Algorithms
- Enhancement of Model Accuracy



The background features three decorative elements consisting of numerous thin, parallel, wavy lines. One element is in the top right corner, another in the bottom left corner, and a third in the bottom right corner. These elements create a sense of movement and depth.

Thank You!



Any doubt/question send me an email: pmapm@ua.pt